MIL-C-47221A(MI)
23 April 1982
Superseding
MIL-C-47221(MI)
12 July 1974

MILITARY SPECIFICATION

COMPOUND, MOLDING, EPOXY RESIN, SILICON DIOXIDE AND

ASBESTOS-FIBER FILLED

This specification is approved for use by the US Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of molding compound consisting of a silicon dioxide, asbestos-fiber filled epoxy resin and a polyamine curing agent.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

* SPECIFICATIONS

* Federal

*	PPP-B-585	BOX, WOOD, WIREBOUND
*	PPP-B-591	BOXES, SHIPPING, FIBERBOARD, WOOD-CLEATED
*	PPP-B-601	BOXES, WOOD, CLEATED PLYWOOD

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: DRSMI-RSDS, Redstone Arsenal, AL 35898, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 9330

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*	PPP-B-621	BOX, WOOD, NAILED AND LOCK CORNER
*	PPP-P-1892	PAINT, VARNISH, LACQUER, AND RELATED MATERIALS, PACKAGING, PACKING AND MARKING OF

STANDARDS

Military

	MIL-STD-129	MARKING FOR SHIPMENT AND STORAGE
*	MIL-STD-1188	COMMERCIAL PACKAGING OF SUPPLIES AND EQUIPMENT 010977

* (Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

American Society for Testing and Materials

ASTM D 115	Varnishes Used for Electrical Insulation, Methods of Testing
ASTM D 229	Rigid Sheet and Plate Materials Used for Electrical Insulation, Method of Testing
ASTM D 638	Tensile Properties of Plastics, Method of Test for
ASTM D 648	Deflection Temperature of Plastics Under Flexural Load, Test Method for
ASTM D 790	Flexural Properties of Plastics and Electrical Insulating Materials, Test Methods for
ASTM D 792	Specific Gravity and Density of Plastics by Displacement, Test Methods for

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ASTM D 1652	Epoxy Content of Epoxy Resins, Methods of Test for
ASTM E 168	General Techniques of Infrared Quantitative Analysis, Recommended Practice for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Uniform Classification Committee

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Blvd, Chicago, IL 60605.)

National Motor Freight Traffic Associations, Inc.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., 1616 P Street, NW, Washington, DC 20036.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

* 3.1 Preproduction sample. Unless otherwise specified (see 6.2), the contractor shall furnish preproduction samples for preproduction inspection in accordance with 4.3. The preproduction samples shall be produced by the same methods, processes and techniques to be employed in subsequent production of the contract quantity.

3.2 Material.

3.2.1 Composition and condition in container.

3.2.1.1 Composition. The compound shall be composed of epoxy resin, asbestos-fiber filled, and amine curing agent.

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3.2.1.2 Condition in container. The material, as received, shall be free of gel particles, dirt, and any contaminants which would affect the intended use.

3.2.1.3 Storage life. The compound shall meet the requirements of this specification after storage in original unopened container for a minimum period of 12 months for the epoxy resin and 6 months for the curing agent, at a temperature of 75 degrees plus or minus 15 degrees Fahrenheit (F) @24 degrees plus or minus 8 degrees Celsius (C)).

3.2.2 Physical and chemical properties. Physical and chemical properties shall be as specified in Table I.

Table I. Physical and chemical properties

Property	Minimum	Maximum
Resin viscosity, 25°C, poise	80.0	100.0
Curing agent viscosity, 25°C, poise	75.0	85.0
Resin weight per gallon, lbs/gal	11.5	12.5
Curing agent weight per gallon, lbs/gal	8.5	9.5
Resin epoxide equivalent weight, grams	290	300
Resin ash content, percent	35.0	38.0
Cured compound ash content, percent	30.0	35.0
Cured compound specified gravity	1.30	1.50

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3.2.3 Curing agent infrared spectrum. The curing agent shall exhibit only the transmittance bands illustrated in the spectrum of figures 1 and 2 within plus or minus 5 percent transmittance.

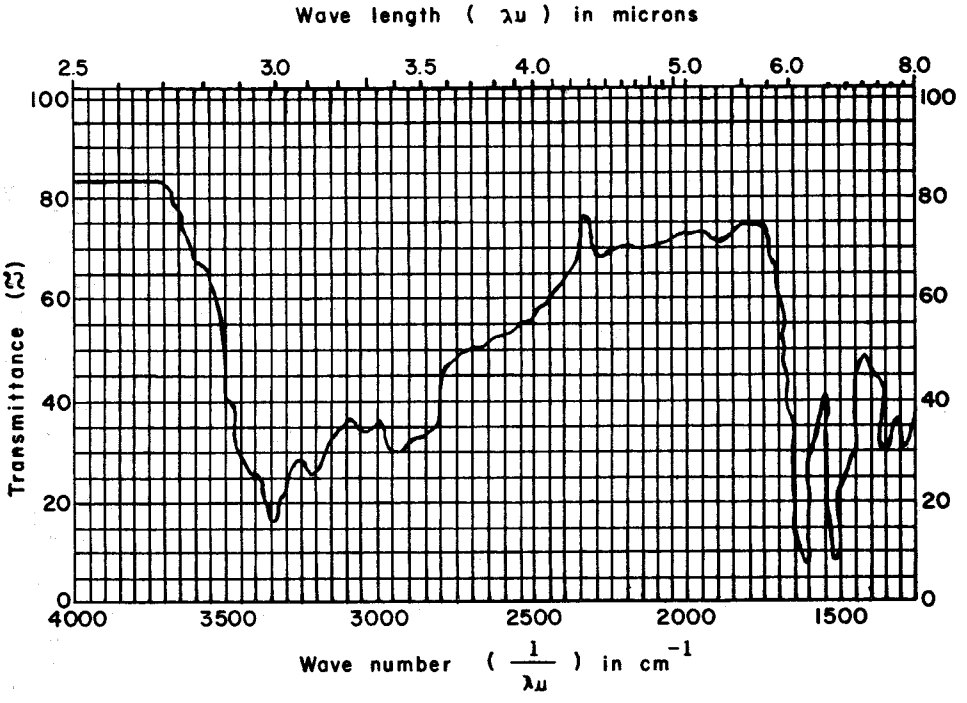


Figure 1. Infrared spectrum from 3700 to 1250 cm^{-1} of the amine curing agent

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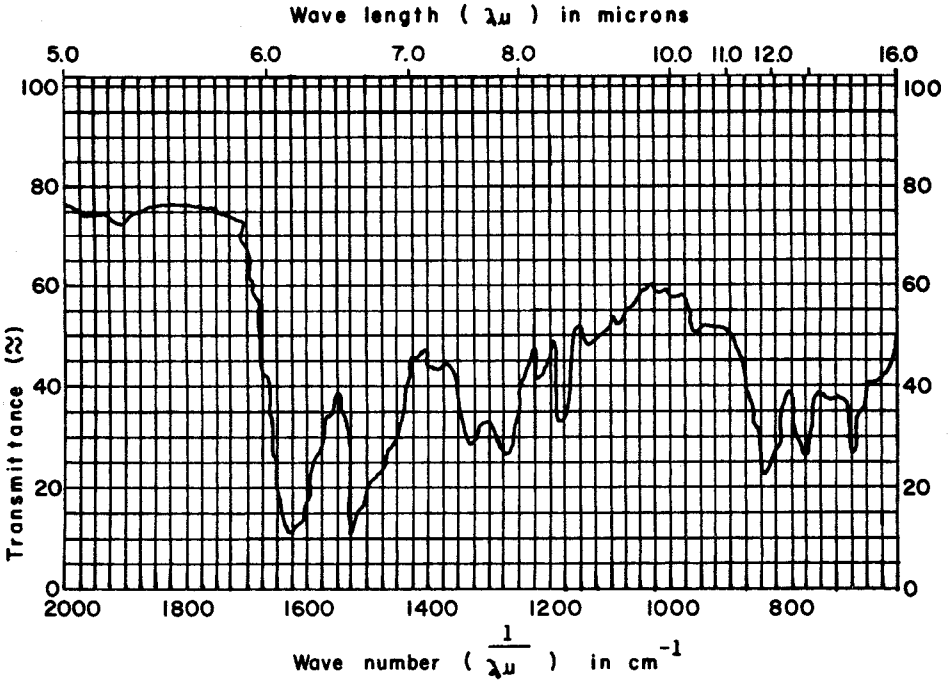


Figure 2. Infrared spectrum from 1700 to 750 cm^{-1} (expanded scale) of the amine curing agent

3.3 Performance characteristics.

3.3.1 Tensile strength. The tensile strength of the cured compound shall be not less than 7,000 pounds per square inch (psi).

3.3.2 Flatwise flexural strength. The flatwise flexural strength shall be not less than 11,000 psi.

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3.4 Environmental conditions.

3.4.1 Heat distortion. The curing compound shall not distort when exposed to a pressure of 264 psi between the temperature of 185 to 195 degrees F (85 degrees to 91 degrees C) and to a pressure of 66 psi between the temperatures of 215 and 225 degrees F (102 degrees to 107 degrees C).

3.5 Workmanship. The material shall be uniform in quality and shall be free from impurities and other defects that could adversely affect its use.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure suppliles and services conform to prescribed requirements.

4.2 Preproduction inspection. When specified in the contractural document (see 6.2), a sample consisting of 5 pounds of resin and 13 ounces of curing agent, manufactured under the same conditions as those proposed for subsequent production inspection shall be as specified herein. If the preproduction sample does not meet the requirements of this specification, it shall be rejected. Subsequent units shall not be considered for acceptance until Government approval of the preproduction sample has been obtained. Units subjected to preproduction inspection shall have successfully passed acceptance inspection. Testing of the preproduction sample, to determine compliance with the characteristics listed in Table II, shall be conducted in accordance with the corresponding test paragraphs.

4.3 Quality conformance inspection.

4.3.1 Lot formation. A lot shall consist of all the material manufactured in one continuous operation by the same process by the same manufacturer with this specification and submitted for inspection at one time.

4.3.2 Visual examination. Visual examination to assure compliance with 3.5 shall be conducted on 100 percent of production.

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4.3.3 Sampling for testing. Sufficient quantities of resin and curing agent to conduct the test specified herein shall be selected at random from one unit representative of each lot. Each container of base resin and curing agent shall be considered as a unit of product. Failure of a sample to meet the quality conformance requirements shall be cause for lot rejection.

Table II. Preproduction inspection

Characteristic	Requirement source	Test paragraph
Composition and condition in container	3.2.1	4.5.4.1
Storage life	3.2.1.3	4.5.4.2
Resin viscosity	Table I	4.5.4.3
Curing agent viscosity	Table I	4.5.4.4
Curing agent weight	Table I	4.5.4.4
Resin epoxide equivalent	Table I	4.5.4.5
Resin ash content	Table I	4.5.4.6
Cured compound ash content	Table I	4.5.4.6
Cured compound specific gravity	Table I	4.5.4.7
Curing agent infrared spectrum	3.2.3	4.5.4.8
Tensile strength	3.3.1	4.5.4.9
Flatwise flexural strength	3.3.2	4.5 4.10
Heat distortion	3.4.1	4.5.4.11
Workmanship	3.5	4.5.5

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4.4 Inspection equipment. The inspection equipment for conducting examination and tests shall be as specified in the applicable test methods and procedures paragraphs.

4.5 Test methods and procedures.

Table III. Acceptance inspection

Characteristic	Requirement source	Test and inspection paragraph
Composition and condition in container	3.2.1	4.5.4.1
Tensile strength	3.3.1	4.5.4.9
Flatwise flexural strength	3.3.2	4.5.4.10
Heat distortion	3.4.1	4.5.4.11
Workmanship	3.5	4.5.5
Packaging, packing, and marking	5.	4.6

4.5.1 Test conditions. Unless otherwise specified herein, the following conditions shall be used as a basis to establish performance requirements.

a. Temperature, room ambient plus or minus 9 degrees C (plus or minus 16 degrees F).

b. Altitude, facility ground.

c. Humidity, facility ambient up to 95 percent relative humidity.

4.5.2 Test sequence. Testing within each classification of inspection (preproduction or acceptance) shall be at the option of the supplier.

4.5.3 Test specimen preparation.

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4.5.3.1 Compound formulation. The compound shall be formulated as specified in Table IV.

4.5.3.2 Casting of specimens. The epoxy resin and the curing agent shall be mixed thoroughly until a homogeneous color is obtained. The mixed components shall be applied in a pourable condition within 1 hour after mixing. Specimens shall be cast to the form and size required by the applicable test methods specified herein.

Table IV. Compound formulation

Ingredients	Parts by weight
Epoxy resin, asbestos-fiber filled	85
Curing agent	15

4.5.3.3 Curing. The molds shall be cured for 2 hours at 140 degrees plus or minus 5 degrees F (60 degrees plus or minus 3 degrees C).

4.5.4 Tests.

4.5.4.1 Composition and condition in container. Composition shall be certified by the supplier and condition in container shall be examined visually to establish compliance with 3.2.1.

4.5.4.2 Storage life. A sufficient amount of the epoxy resin and of the curing agent in original unopened containers shall be stored under the conditions specified in 3.2.1.3. At the end of the storage period, each component and the compound shall be subjected to all the tests specified herein. The supplier's certification of compliance certifying to the storage life requirements (see 3.2.1.3) may be accepted during the storage period.

4.5.4.3 Viscosity test. The viscosity of the asbestos-fiber filled epoxy resin and the polyamine curing agent shall be determined in accordance with ASTM D 115, and shall conform to Table I.

4.5.4.4 Weight per gallon. The weight per gallon of the asbestos-fiber filled epoxy resin and the polyamine curing agent shall be determined in accordance with ASTM D 792, and shall conform to Table I.

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4.5.4.5 Epoxide equivalent weight. The epoxide equivalent weight of the asbestos-fiber filled epoxy resin shall be determined in accordance with ASTM D 1652, and shall conform to Table I.

4.5.4.6 Ash content. The ash content of the asbestos-fiber filled epoxy resin and the cured compound shall be determined in accordance with ASTM D 229, and shall conform to Table I.

4.5.4.7 Specific gravity. The specified gravity of the cured compound shall be determined in accordance with ASTM D 972, and shall conform to Table I.

4.5.4.8 Curing agent infrared spectrum. The infrared transmittance spectrum of the curing agent shall be determined in accordance with ASTM E 168, and shall conform to 3.2.3.

4.5.4.9 Tensile strength. The tensile strength of the curing compound shall be determined in accordance with ASTM D 638, and shall conform to 3.3.1.

4.5.4.10 Flatwise flexural strength. The flatwise flexural strength of the cured compound shall be determined in accordance with ASTM D 790, and shall conform to 3.3.2.

4.5.4.11 Heat distortion temperature. The heat distortion temperature of the cured compound shall be determined in accordance with ASTM D 648, and shall conform to 3.4.1.

4.5.5 Workmanship. The material shall be visually examined to assure conformance to the workmanship requirements specified in 3.5.

4.6 Packaging, packing, and marking. The inspector shall ascertain that the packaging, packing, and marking of the material conforms to this specification.

5. PREPARATION FOR DELIVERY

* 5.1 Preservation, packaging, packing, unitilization and marking. Level A, B or Industrial (see 6.2).

* 5.1.1 Level A or B. The compound shall be packaged in accordance with requirements of PPP-P-1892.

* 5.1.2 Industrial. Compound shall be packaged in accordance with requirements of MIL-STD-1188.

* 5.2 Packing.

* 5.2.1 Level A. The compound shall be packed in overseas type exterior containers conforming to one of the following: PPP-B-585, PPP-B-591, PPP-B-601 or PPP-B-621.

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* 5.2.2 Level B. Same as Level A except domestic type exterior containers shall be used.

* 5.2.3 Industrial. Packing shall be in accordance with requirements of MIL-STD-1188.

* 5.3 Marking.

* 5.3.1 Level A or B. Marking shall be in accordance with requirements of MIL-STD-129.

* 5.3.2 Industrial. Marking shall be in accordance with requirements of MIL-STD-1188.

* 5.3.3 Special marking. Shelf-life markings may be required (see 3.2.1.3).

6. NOTES

6.1 Intended use. This compound is intended for use where relatively high temperature resistance is required, as in holding fixtures for arc-lamps and, when metallized, in reflectors for different light sources.

* 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

* b. Whether a preproduction sample is required (see 3.1 and 4.2), and if so, pertinent details.

c. Requirements for certification of composition (see 4.5.4.1).

d. Size of container.

* e. Level of packaging required (see 5.1).

* 6.3 Changes to previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modification, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER AND TITLE MIL-C-47221A(MI) COMPOUND, MOLDING, EPOXY RESIN, SILICON DIOXIDE AND ASBESTOS-FIBER FILLED

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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